



École Nationale Supérieure d'Intelligence Artificielle
ENSIA

Comprehensive Data Mining Analysis Report Algerian Laptop Market

Price Prediction and Market Insights

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1 Introduction

1.1 Objectives

The primary objectives of this study are:

1. **Predictive Modeling:** Develop accurate price prediction models using regression techniques
2. **Clustering Modeling:** Group similar laptops based on their features to identify market segments and pricing patterns.
3. **Association Rules:** Assessing whether Association Rules mining techniques could be used to extract hidden meaningful patterns.
4. **Feature Engineering:** Create meaningful features that capture hardware performance, build quality, and market dynamics
5. **Business Intelligence:** Extract actionable insights for both buyers and sellers

1.2 Methodology Overview

1. **Business Understanding:** Define objectives from buyer and seller perspectives
2. **Data Understanding:** Assess data quality, completeness, and characteristics
3. **Data Preparation:** Clean, transform, and engineer features
4. **Modeling:** Build and compare multiple regression models, and run clustering algorithms
5. **Evaluation:** Assess model performance using appropriate metrics
6. **Deployment:** Extract and communicate business insights

2 Business Objectives

Understanding the distinct needs of buyers and sellers is fundamental to developing a model that serves both stakeholder groups effectively.

2.1 Buyer Perspective

From a buyer's perspective, the primary objectives are:

- **Value Maximization** Buyers seek to identify the optimal laptop configuration within their budget constraints. This requires understanding which features provide the best price-to-performance ratio in the Algerian market context.
- **Price Optimization** Buyers need to identify fair market prices and avoid overpaying. This requires understanding price distributions, typical premiums for specific features, and temporal price variations.

- **Feature Prioritization** Given budget constraints, buyers must understand which features justify premium pricing. For example, is a newer generation CPU worth more than additional RAM? Does a dedicated GPU provide sufficient value for non-gaming use cases?
- **Condition Assessment** The secondary market includes both new and used laptops. Buyers need to quantify the appropriate discount for used items to make informed decisions.
- **Timing Strategy** Identifying when median prices are lower helps buyers time purchases to maximize value.

2.2 Seller Perspective

From a seller’s perspective, the objectives are complementary:

- **Competitive Pricing** Sellers must price laptops competitively to maximize sales velocity while avoiding underpricing. Understanding market price distributions and competitor pricing is essential.
- **Feature Highlighting** Sellers need to understand which features command premium prices to effectively market their listings. This knowledge enables strategic feature emphasis in listings.
- **Condition Pricing** Sellers must apply appropriate depreciation for used items. Overestimating depreciation reduces revenue; underestimating reduces competitiveness.
- **Timing Strategy** Knowing when listing volume peaks and whether median price rises or falls with volume helps sellers time listings for visibility and optimal pricing.
- **Brand Positioning** Understanding brand premiums (e.g., Apple vs. HP vs. Lenovo for similar specs) allows sellers to set competitive prices and position listings effectively.

3 Data Assessment

3.1 Dataset Overview

The dataset consists of 16,283 laptop listings collected from Ouedkniss, containing 14 features describing hardware specifications, condition, location, and temporal information.

Table 1: Dataset Structure

Attribute	Count	Non-null (%)
price_preview	16,331	99.5%
created_at	16,406	100.0%
city	16,333	99.6%
spec_Etat (Condition)	9,600	58.5%
model_name	15,878	96.8%
DEDICATED_GPU	3,679	22.4%
CPU	16,406	100.0%
RAM_SIZE	15,952	97.2%
SSD_SIZE	14,498	88.4%
HDD_SIZE	1,166	7.1%
SCREEN_SIZE	14,256	86.9%
SCREEN_FREQUENCY	1,053	6.4%
SCREEN_RESOLUTION	6,149	37.5%
RAM_TYPE	5,347	32.6%

3.2 Data Quality Assessment

3.2.1 Missing Values Analysis

The dataset exhibits significant missingness in several features:

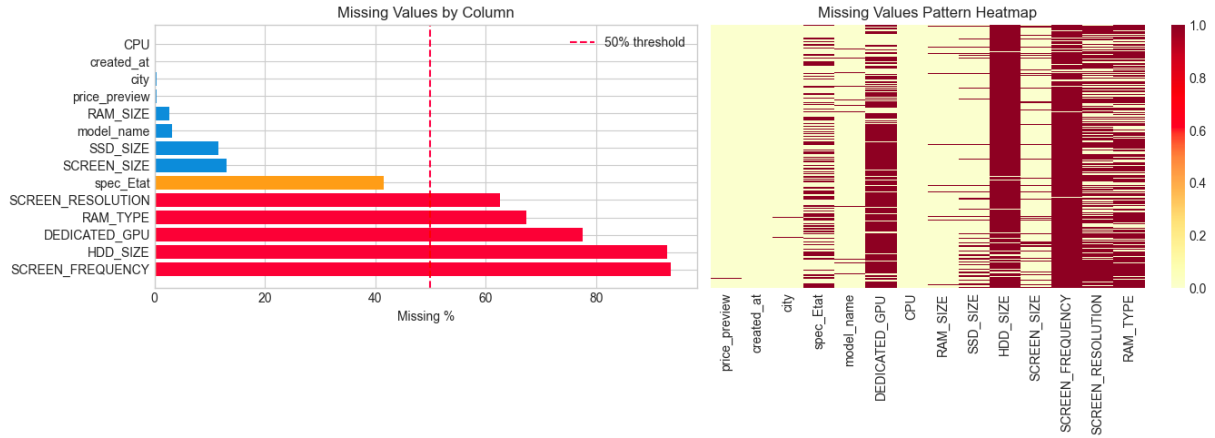


Figure 1: Missing Values Bar Chart and Heatmap Pattern

The missingness pattern reveals domain-specific characteristics:

- GPU information is often omitted for non-gaming laptops (expected behavior, as they most likely don't have dedicated GPUs)
- RAM_TYPE missingness is high, requiring inference from CPU generation (domain knowledge)

3.2.2 Price Distribution Analysis

Initial price analysis reveals significant data quality issues requiring correction:

- Prices range from near-zero (e.g., 1 DZD) to over 75 million DZD, indicating data entry errors
- Distribution is heavily right-skewed (skewness > 2)
- Multiple price anomalies suggest unit confusion (centimes vs. DZD) and missing zeros

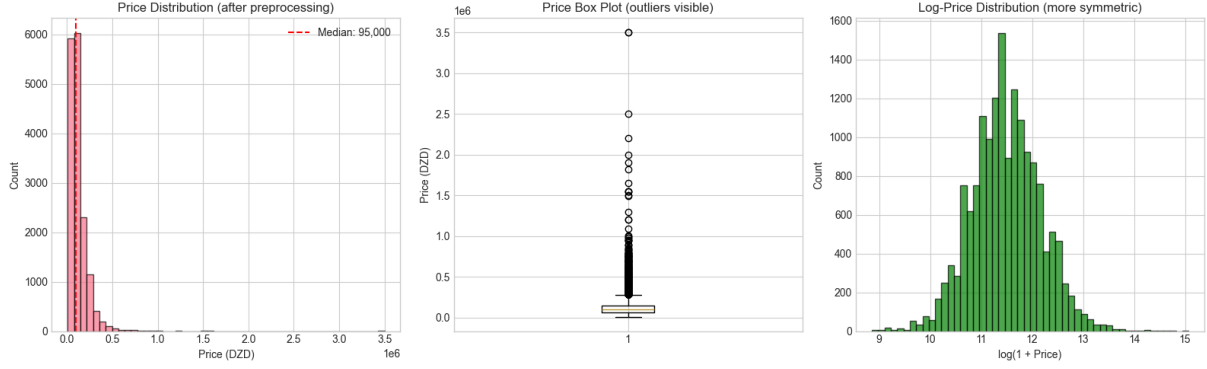


Figure 2: Initial Price Distribution (Histogram, Box Plot, Log-Scale)

3.3 Data Characteristics

3.3.1 Temporal Distribution

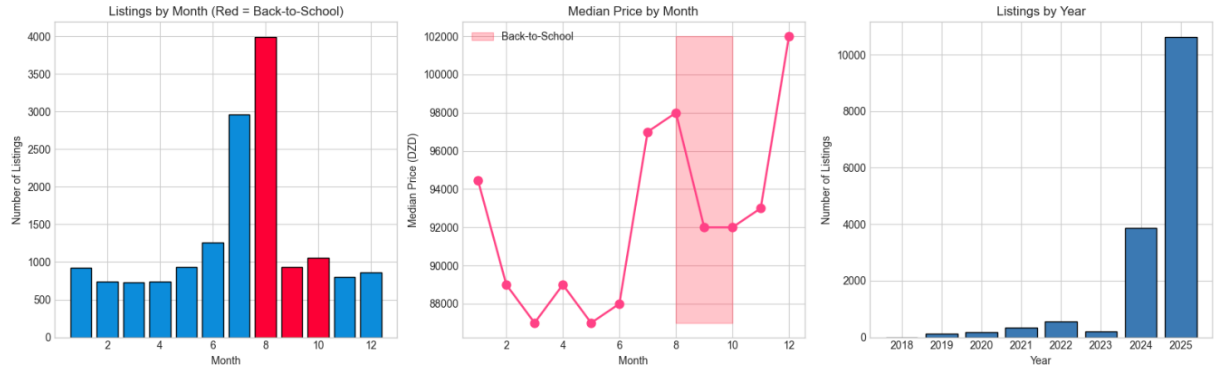


Figure 3: Temporal Distribution of Listings

3.4 Data Quality Summary

The dataset presents both opportunities and challenges:

Strengths:

- Large sample size (16,406 listings)
- Core features (CPU, RAM, Storage) highly complete
- Temporal information available for time-series analysis

- Price information nearly complete

Challenges:

- Significant missingness in GPU and display features
- Price data requires extensive cleaning
- Model names require parsing and standardization
- Missing RAM_TYPE requires inference from CPU generation

4 Data Cleaning and Pre-Processing

4.1 Price Cleaning Strategy

Price data exhibited multiple anomalies requiring correction:

4.1.1 Anomaly Detection

Three categories of price anomalies were identified:

1. **Missing Zeros:** Prices < 100 DZD(e.g., 75 instead of 75,000)
2. **Unit Confusion:** Prices $> 2,000,000$ DZD(centimes entered as DZD)
3. **Unrealistic Values:** Prices $< 15,000$ DZDor $> 800,000$ DZD(outside considerably most of market range values)

4.1.2 Correction Methodology

- **Missing Zero Correction:** For prices < 100 DZD, we multiply by 1,000 if the listing contains valid specifications (CPU or RAM), indicating a data entry error rather than a legitimate low price.
- **Unit Confusion Correction:** Prices $> 2,000,000$ DZDare divided by 100, assuming centimes were entered as DZD.

4.2 Pre-Processing Summary

Table 2: Data Cleaning Summary

Step	Removed/Corrected	Remaining
Initial Dataset	–	16,406
Preprocessing (price cleaning, filtering)	123 removed	16,283
Temporal Train/Test Split (80/20)	–	13,026 / 3,257
IQR Outlier Removal (train-set bounds)	117 removed	12,916 / 3,250

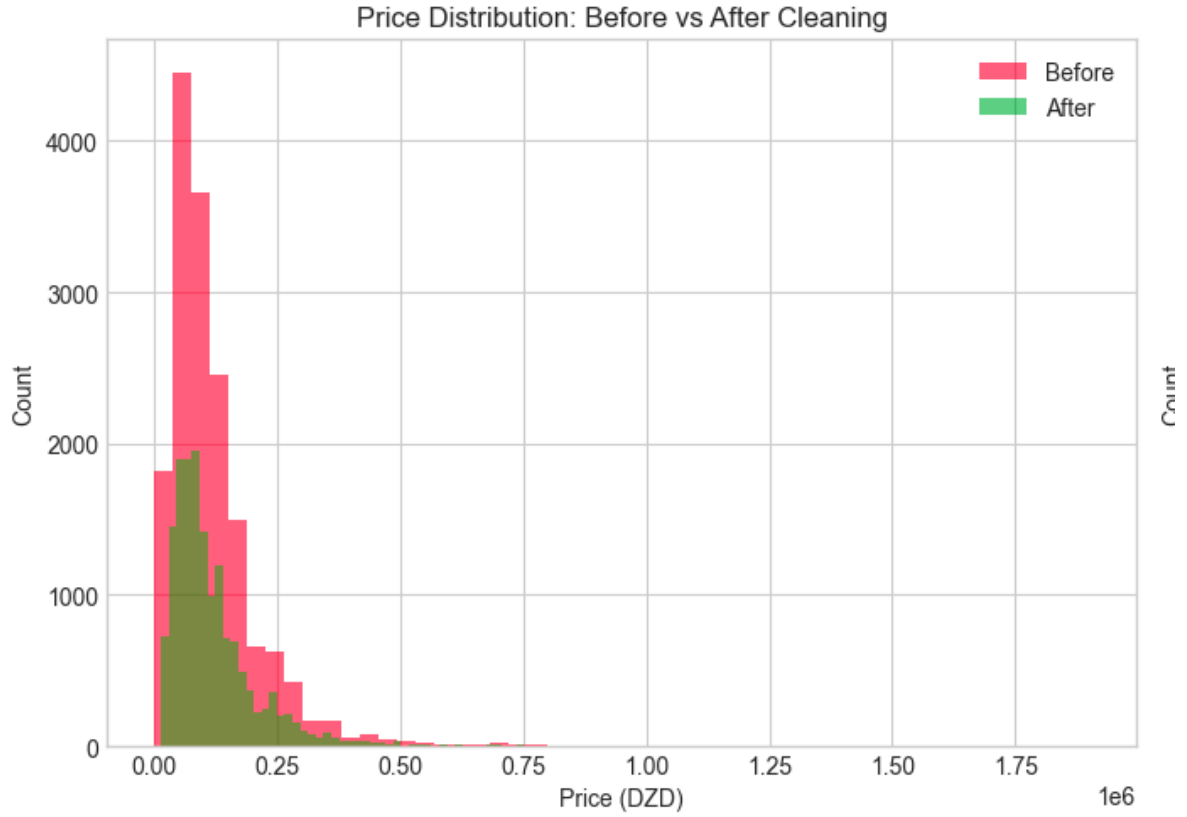


Figure 4: Price Distribution Before and After Cleaning

5 Feature Engineering

This section details the creation of the engineered features used for modeling (including the segmentation-derived cluster label).

5.1 Brand and Series Extraction

5.1.1 Motivation

Laptop series (e.g., ThinkPad, XPS, ROG) are stronger price indicators than brand alone, as they reflect build quality, target market, and feature sets.

5.1.2 Extraction Methodology

We implement keyword-based extraction from `model_name`:

1. **Brand Detection:** Map keywords (e.g., "LENOVO", "THINKPAD") to brands
2. **Series Detection:** Identify series (e.g., "LEGION", "XPS", "OMEN")
3. **Build Quality Tier:** Map series to ordinal tiers (1–5):
 - Tier 1: Budget/Stream (e.g., HP Stream)
 - Tier 2: Consumer (e.g., Inspiron, IdeaPad)
 - Tier 3: Business (e.g., ProBook, ThinkBook)

- Tier 4: Premium (e.g., XPS, ThinkPad, ZenBook)
- Tier 5: Workstation/Gaming (e.g., ZBook, ROG, Legion, Precision)



Figure 5: Brand Distribution and Price by Brand

5.2 CPU Feature Engineering

From the CPU field, we extract:

- **cpu_brand**: Intel, AMD, Apple, or Unknown
- **cpu_family**: 3, 5, 7...
- **cpu_generation**: Extracted via regex (e.g., "10th Gen", "i7-12700H" → 12)
- **cpu_suffix**: Power class (U, P, H, HK, HX)
- **is_pro**: Boolean flag indicating if the CPU is a "Pro" variant
- **cpu_performance_class**: A heuristic classification into 4 performance tiers based on brand, family, generation, and suffix (High, Mid-Low...)
- **inferred_ddr** and **inferred_ddr_ordinal**

5.2.1 PassMark CPU: Hierarchical Matching and Multi-Dimensional Features

We enrich CPU with **PassMark** data from `cpubenchmark.net`, exported to `cpu_passmark.csv` (name, cpumark, single_thread, cores, threads) via `export_passmark_data.py`. To avoid wrong matches (e.g. “i5 12th” matching an i3), we use **hierarchical matching**:

1. **Level 1 — Exact model match**: Normalize the listing CPU to a canonical token (e.g. i7-11800H → i711800h) and look it up in a PassMark index.
2. **Level 2 — Fuzzy match within brand/tier**: Restrict candidates to the same brand and tier (e.g. Intel i7), then best match by word overlap.
3. **Level 3 — Fallback**: Use the **median** of cpumark, single_thread, cores, and threads over the (Brand + Tier + Generation) group in PassMark.

From the matched (or fallback) PassMark row we inject **multi-dimensional CPU features** used in regression and clustering:

- **cpu_mark**: Multi-thread score (throughput).
- **cpu_single_thread**: Single-thread score (responsiveness, games, UI).
- **cpu_cores**, **cpu_threads**: Core and thread counts (parallelism, segmentation).

5.2.2 Inferred DDR Generation

Given high missingness in `RAM_TYPE` (67.4%), we infer DDR generation from CPU brand and generation (domain-knowledge, it may have some errors, but in general it is as the following)

Table 3: Inferred DDR Generation from CPU Characteristics (Domain Knowledge)

CPU Brand	Generation / Model	Suffix	Inferred DDR
Intel	Gen 1–7	any	DDR3
Intel	Gen 8–11	any	DDR4
Intel	Gen 12+	U / P	LPDDR5
Intel	Gen 12+	H / HX	DDR5
AMD Ryzen	1000–5000 series	any	DDR4
AMD Ryzen	6000 series	any	LPDDR5
AMD Ryzen	7000+ series	U / HS	LPDDR5
AMD Ryzen	7000+ series	H / HX	DDR5
Apple	M1	–	LPDDR4X
Apple	M2	–	LPDDR5
Apple	M3 / M4	–	LPDDR5X
Unknown / Other	–	–	Unknown (0)

5.3 GPU Feature Engineering

5.3.1 Extracted Features

From `DEDICATED_GPU`, we extract:

- **has_gpu**: Binary flag (1 if dedicated GPU present)
- **gpu_brand**: NVIDIA, AMD, INTEL or NONE
- **gpu_tier**: The specific model name extracted (e.g., RTX 3060, GTX 1650, RTX 4090..)
- **gpu**: Performance or form-factor modifiers (e.g., Ti, MAX-Q, SUPER)
- **gpu_performance_class**: High, Low..

5.3.2 GPU Benchmark: PassMark 3D and Tier Fallback

GPU score comes from one source: **PassMark 3D** (videocardbenchmark.net), pre-exported to `gpu_passmark.csv` via `export_passmark_data.py`. For each listing with a dedicated GPU we match the GPU text to PassMark (word-overlap); on match we assign the g3d score to `gpu_score`. So `gpu_score` is a single, comparable column: either the matched PassMark g3d or the tier fallback.

5.3.3 Tier Fallback (No Match or CSV Missing)

When no PassMark match is found (or the CSV is missing), we assign **tier-based fallbacks** (realistic PassMark 3D median value per tier, guided by GPU performance categorical variable) so every dedicated-GPU listing has a valid `gpu_score`. Listings without a dedicated GPU receive `gpu_score = 0`.

5.4 Storage and Display Features

5.4.1 Storage Performance Score

We define `storage_perf_score` (0/1/2) to capture storage type impact on perceived value and performance: 0 = no storage specified, 1 = HDD only, 2 = SSD present.

5.4.2 Screen Resolution and Pixel Count

From `SCREEN_RESOLUTION` strings we parse width, height, and total pixel count (`screen_w`, `screen_h`, `screen_pixels`) and derive a `resolution_class` (e.g., FHD, QHD). Pixel count and class are used in segmentation and regression so that display sharpness and panel tier can inform price prediction where resolution is available.

5.5 Wilaya and Region (Geographic Features)

The `city` field is mapped to Algeria's 58 **wilayas** (administrative regions) via a domain-defined city-to-wilaya dictionary. Wilayas are then grouped into three **regions** for regional pricing and concentration analysis:

- **North Coast**: Coastal wilayas (e.g., Alger, Oran, Béjaia)
- **Highlands**: Inland wilayas (e.g., Sétif, Batna)
- **South**: Sahara and southern wilayas (e.g., Ouargla, Biskra, Tamanrasset)

Binary flags `is_alger` (Alger wilaya) and `is_north_coast` are also created. These features support **regional analysis** in EDA (market concentration, price variation by region) and can be used in modeling to capture geographic price effects.

5.6 Condition Encoding

Based on Algerian market knowledge, we encode condition with value retention weights:

Table 4: Condition Value Retention (Algerian Market)

Condition	Value Retention	Depreciation
Jamais utilisé (New/Sealed)	1.00	0%
Bon état (Good)	0.75	25%
Moyen (Fair)	0.60	40%
Unknown	0.70	30%

5.7 Market Segmentation (K-Means)

To capture latent market segments and improve price prediction, we cluster laptops by **configuration only** (no price), then use the cluster label as a categorical feature in regression. This choice is justified by the fact that pricing patterns differ by segment (e.g., GPU matters more in gaming than in office laptops).

The train/test split is applied first; missing values in the clustering features are imputed using **training-set medians only** (same rule as in the regression pipeline). K-Means is fitted on the **training set only** (on scaled configuration features), and K is chosen via an **elbow plot** (within-cluster SSE vs. K). Test listings are assigned to the nearest cluster centroid, so no future information is used and there is no data leakage. The resulting `cluster_id` is added to the regression feature set, allowing the model to learn segment-specific pricing.

5.7.1 Interpretation of the Segments

With $K = 4$, the clusters obtained from configuration features (price excluded) can be summarized as interpretable market segments. EDA analysis reveals the following cluster profiles:

Table 5: K-Means Market Segments (EDA Results)

Segment	Count	Market Share	Median Price	RAM	SSD
Budget (Cluster 0)	3,326	27.5%	75,000 DZD	8 GB	256 GB
Mid-Range (Cluster 2)	5,627	46.6%	85,000 DZD	16 GB	256 GB
Premium (Cluster 1)	1,653	13.7%	132,000 DZD	16 GB	512 GB
High-End/Gaming (Cluster 3)	1,473	12.2%	240,000 DZD	16 GB	1 TB

Key observations:

- The **Mid-Range** segment dominates the market (46.6% of listings), reflecting typical consumer demand.

- **Budget** laptops (older CPUs, 8 GB RAM) constitute 27.5% of the market.
- **Premium** and **High-End/Gaming** segments together represent about 26% of listings but command significantly higher prices.
- Storage capacity clearly differentiates segments: 256 GB (Budget/Mid-Range) → 512 GB (Premium) → 1 TB (High-End).

5.7.2 Dual Use: Regression and Market Insight

The same K-Means segmentation is used for two purposes. **First**, `cluster_id` is included as a categorical feature in the regression model so that the relationship between features and price can differ by segment (e.g., GPU score has a stronger marginal effect in the gaming segment than in the office segment) since the price drivers are not uniform across the market.

Second, the segments serve as a **market structure** summary for reporting and business insight: we can describe each segment by size, typical price range, and typical specs, and tailor recommendations for buyers and sellers by segment.

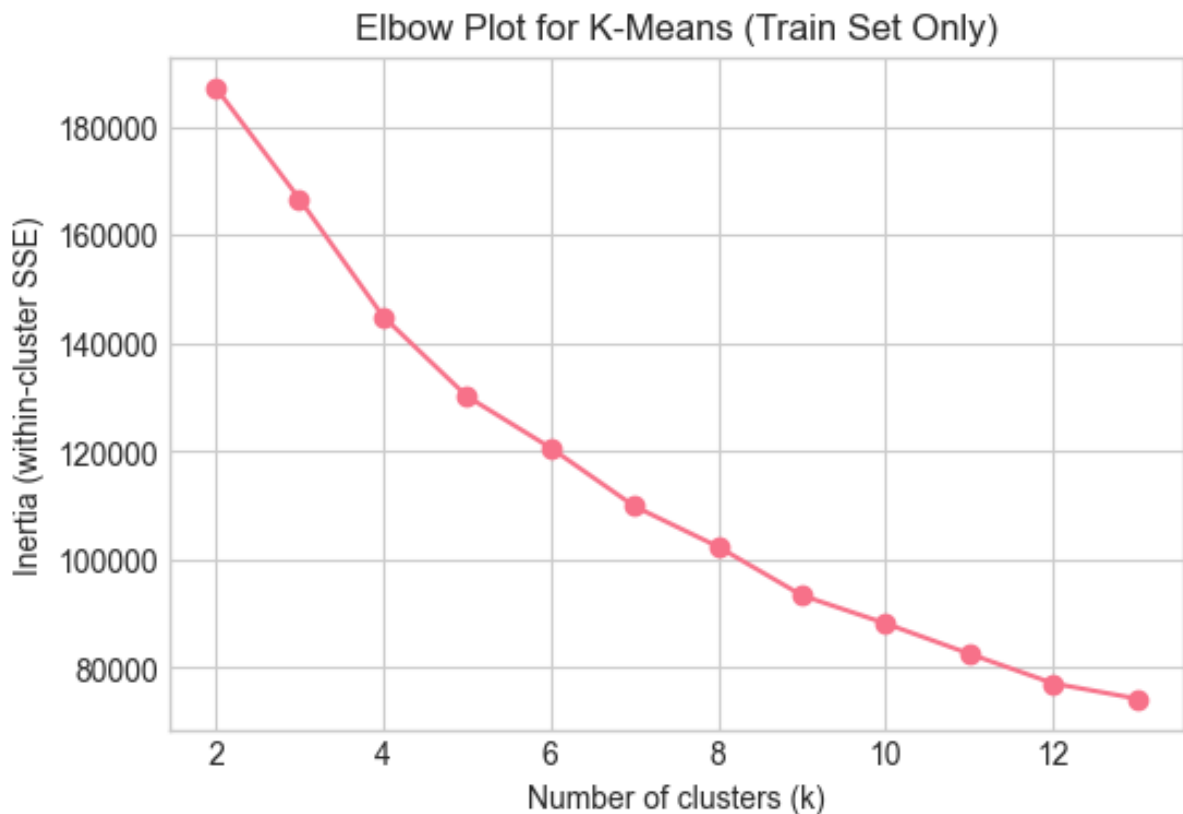


Figure 6: Elbow Plot for K-Means (choice of number of clusters)

5.8 Exploratory Data Analysis and Key Visualizations

The EDA notebook produces several visualizations and analyses that inform modeling and business insights. Key additions are summarized below.

5.8.1 Regional Analysis (Wilaya Distribution)

- **Market concentration:** The top three wilayas (typically Alger, Oran, Blida) account for approximately **76.7%** of all listings; indicating a highly concentrated market in major urban areas.
- **Regional grouping:** Wilayas are grouped into North Coast, Highlands, and South. Listings are dominated by North Coast; median price is compared by region (% vs. overall median) to quantify regional premiums or discounts.

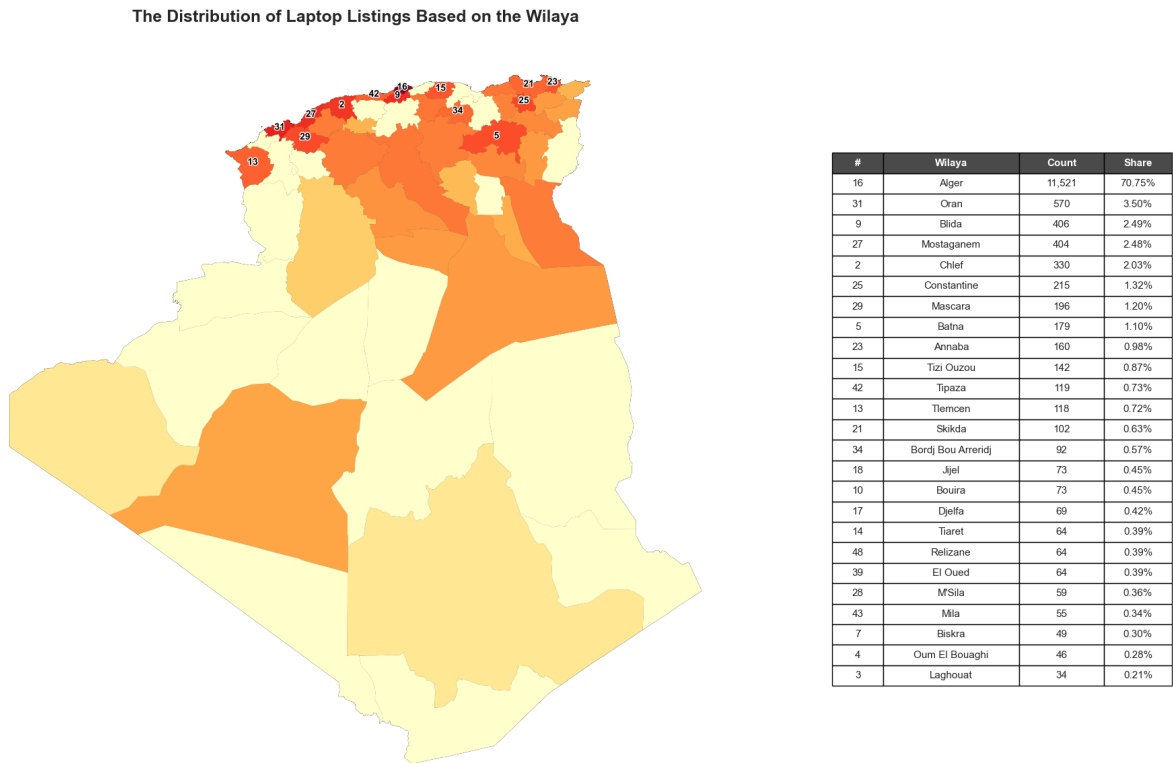


Figure 7: Regional Analysis: Top Wilayas by Listing Count, Market Share by Region, and Price Distribution by Region.

5.8.2 Brand Premiums and Gaming Series Premium

We quantify **brand premiums** in two ways: (1) **Raw**: median price by brand (e.g., Apple vs. Dell) across all listings; (2) **Similar specs**: median price by brand within matched hardware segments (same CPU performance class, GPU class, and condition) to control for configuration. Key findings from EDA:

- **Apple vs Dell (raw)**: Apple commands +71% premium (Apple median: 149,000 DZD, Dell: 87,000 DZD)
- **Apple vs Dell (similar specs)**: +44% average across 14 comparable segments
- **Apple vs Lenovo**: +82% raw premium
- **Apple vs HP**: +62% raw premium

We also compute **gaming series vs. consumer series** premium:

- **Raw:** Gaming series = +130% vs. consumer series (Gaming median: 205,000 DZD, n=1,381; Consumer median: 89,000 DZD, n=14,902)
- **Similar specs** (same CPU class + condition): Gaming series = +63% vs. consumer (average over 10 segments)

These findings confirm that both brand positioning and gaming segment carry significant price premiums beyond raw specifications.

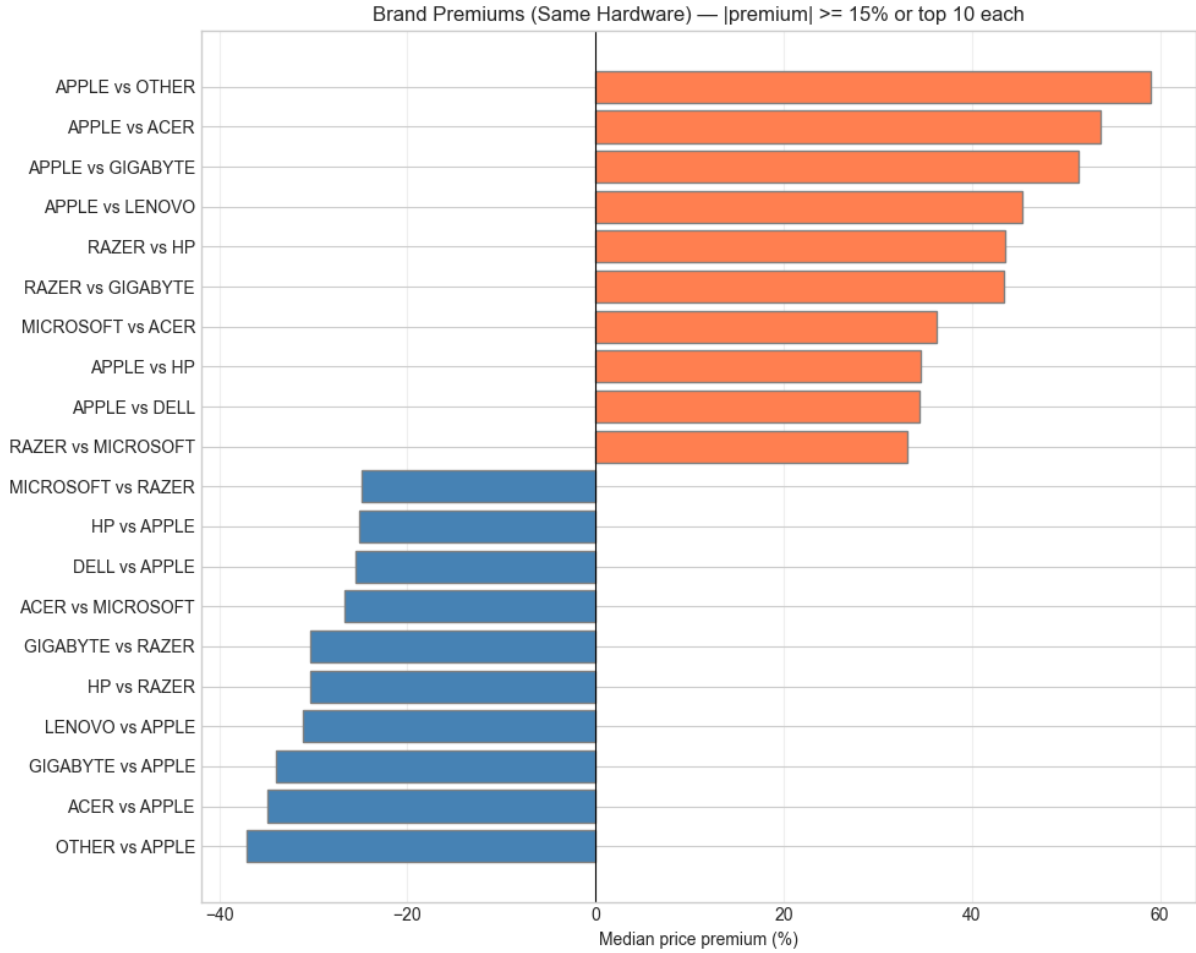


Figure 8: Brand Premiums (same hardware segments) and Gaming vs. Consumer series premium (see EDA notebook).

6 Modeling and Performance

6.1 Model Selection Strategy

We evaluate multiple regression algorithms to identify the best-performing model: Linear Regression, Ridge Regression, Lasso, Random Forest, Gradient Boosting, and SVR. The target variable is log-transformed price: $y_{\log} = \log(1 + \text{price})$.

6.1.1 Feature Set

The model uses 27 raw features (before one-hot encoding), organized into three groups:

- **Numeric features (13):** RAM_SIZE_GB, SSD_SIZE_GB, cpu_mark, cpu_single_thread, gpu_score, build_quality_tier, condition_value_retention, cpu_generation, storage_perf_score, cpu_cores, inferred_ddr_ordinal, SCREEN_SIZE_NUM, cpu_threads
- **Binary features (3):** is_pro, is_gaming_series, has_gpu
- **Categorical features (11):** series, cpu_family, gpu_tier, condition, gpu_suffix, cpu_suffix, cpu_gen_brand, listing_year, brand, ram_type_class, resolution_class

After one-hot encoding of categorical features, the total feature count is 192. Note that `cluster_id` from K-Means segmentation is evaluated separately in the enhanced model comparison.

6.2 Validation Strategy: Time-Series Split

6.2.1 Motivation

Standard K-fold cross-validation violates temporal ordering, potentially causing data leakage (future information predicting past prices). We implement **time-series split**:

- **Training Set:** First 80% of data (chronologically earliest)
- **Test Set:** Last 20% of data (chronologically most recent)

This simulates real-world deployment where models predict future prices based on historical data. From a **business perspective**, buyers and sellers care about how well the model will perform on *new* listings; evaluating on the most recent 20% of listings answers that question directly and avoids overstating performance.

6.3 Imputation of Missing Values

Missing values in the feature set are imputed **after** the train/test split. We compute the median of each feature on the **training set only**, then fill missing values in both training and test sets with these training medians. The same rule is used in the segmentation pipeline: split first, then impute using training-set medians only. This avoids data leakage (test or future information influencing the inputs to the model).

6.4 Evaluation Metrics

We employ three complementary metrics:

6.4.1 R^2 Score (Coefficient of Determination)

Measures proportion of variance explained:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (1)$$

6.4.2 Mean Absolute Error (MAE)

Average absolute prediction error in DZD:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

6.4.3 Mean Absolute Percentage Error (MAPE)

Percentage-based error, interpretable across price ranges:

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3)$$

Why MAPE?

- Very high variance of prices range
- Absolute errors (MAE) are less interpretable across this range
- MAPE provides percentage error (e.g., "15% error" is meaningful for both 50K and 500K laptops)
- Business stakeholders understand percentage errors intuitively

6.5 Model Performance Results

Table 6: Model Performance Comparison (Temporal Test Set)

Model	R ²	MAE (DZD)	RMSE (DZD)	MAPE (%)
Linear Regression	0.8487	21,745	39,174	18.28
Ridge Regression	0.8586	21,630	37,876	18.22
Lasso	0.7567	29,191	49,675	23.50
Random Forest	0.8882	19,205	33,670	16.79
Gradient Boosting	0.9013	17,678	31,638	15.17
SVR	0.8879	19,194	33,721	16.36

6.6 Best Model: Gradient Boosting

Gradient Boosting Regressor achieves the best test performance with:

- R² Score: 0.9013 (90.1% of price variance explained on the test set)
- MAPE: 15.17% (interpretable as average percentage error)
- MAE: 17,678 DZD
- RMSE: 31,638 DZD

Learning curves and residual diagnostics confirm that the model (Gradient Boost and SVR) generalize well, and Linear Regression underfitted; which is expected since it fails to capture non-linear interactions. Note that adding `cluster_id` from K-Means segmentation did not improve Gradient Boosting performance (R^2 decreased slightly from 0.9013 to 0.8969), since Gradient Boosting can already learn non-linear interactions. However, Linear Regression with `cluster_id` achieved R^2 of 0.8505, showing that segments help simpler models.

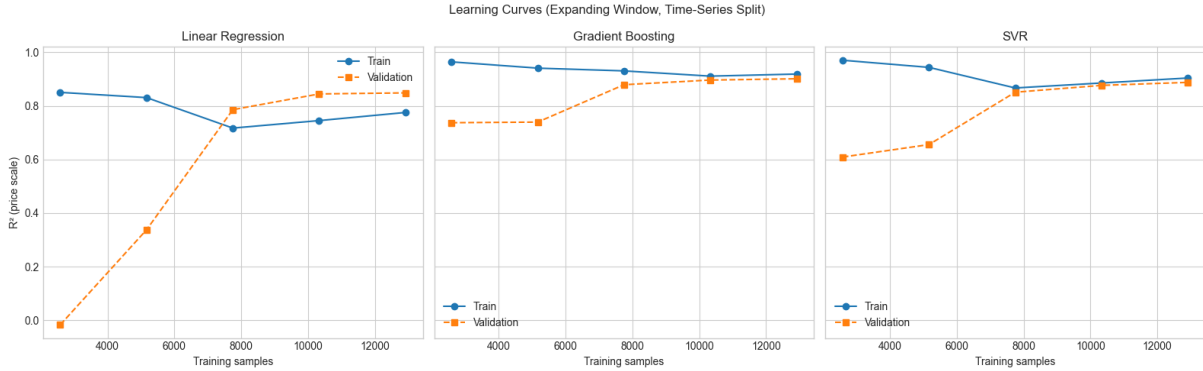


Figure 9: Learning curve

6.7 Feature Importance Analysis

The Gradient Boosting model reveals which features most strongly influence laptop prices. After one-hot encoding categorical features, the model uses 192 total features. The top 10 most important features are:

Table 7: Top 10 Feature Importances (Gradient Boosting)

Feature	Importance
cpu_mark (PassMark multi-thread)	0.37
cpu_single_thread (PassMark)	0.09
SSD_SIZE_GB	0.07
RAM_SIZE_GB	0.06
inferred_ddr_ordinal	0.06
cpu_gen_brand_APPLE_0	0.05
gpu_score	0.04
cpu_generation	0.03
condition_value_retention	0.02
cpu_cores	0.02

Key Insights:

- **CPU dominates pricing:** `cpu_mark` alone accounts for 37% of the model's predictive power, confirming that CPU performance is the primary price driver.
- **Storage and RAM matter:** SSD size and RAM size together contribute 13% of importance, highlighting the value of storage and memory upgrades.

- **DDR generation signals value:** The inferred DDR ordinal (DDR3 < DDR4 < DDR5) captures platform modernity.
- **Apple commands premiums:** The Apple brand/generation interaction feature appears in the top 6, reflecting Apple’s distinct pricing.
- **GPU matters less than expected:** GPU score importance (0.04) is lower than CPU, suggesting that for the general market, CPU outweighs GPU in price determination.

6.8 Model Limitations

- **High-price prediction accuracy:** Limited training data for laptops > 600,000 DZD reduces accuracy at the high end.
- **Feature interactions:** Some complex interactions (e.g., CPU-GPU combinations) may not be fully captured.
- **Temporal trends:** Model assumes stable relationships; market shifts over time may require retraining.
- **Missing feature impact:** High missingness in some features (e.g., screen resolution) limits their use.

7 Business Insights

This section synthesizes actionable insights for both buyers and sellers, derived from model analysis and feature importance.

7.1 Insights for Buyers

Buyers should prioritize CPU performance when comparing options. The feature importance analysis reveals that `cpu_mark` (PassMark multi-thread score) is by far the most important price driver (importance = 0.37), followed by `cpu_single_thread` (0.09), `SSD_SIZE_GB` (0.07), `RAM_SIZE_GB` (0.06), and `inferred_ddr_ordinal` (0.06). This confirms that CPU performance significantly impacts market pricing.

7.1.1 GPU Value Assessment

Buyers should assess whether they need a dedicated GPU:

- **Gaming/Content Creation:** GPU is essential; expect 50–150K DZD premium
- **General Use/Office:** Integrated graphics sufficient; save 50–150K DZD

Actionable Advice:

- Non-gamers: Prioritize CPU and RAM over GPU to maximize value
- Gamers: RTX 30/40 series provide best performance-to-price ratio
- Avoid gaming series laptops without GPUs (likely errors or poor value)

7.1.2 Condition-Based Discounts

Key Finding: EDA analysis reveals significant condition-based price differences:

- **JAMAIS UTILISÉ (New/Sealed):** Median price 159,000 DZD
- **BON ÉTAT (Good condition):** Median price 75,000 DZD
- **Premium:** New laptops command +112% over good condition

Implication: Buyers can achieve significant savings (approximately 50%) by accepting used laptops in good condition.

Actionable Advice:

- “BON ÉTAT” laptops offer substantial savings with minimal risk
- For budget-conscious buyers, condition is a major lever for value optimization
- Unknown condition: Negotiate based on visible condition and market comparisons

7.1.3 RAM and Storage Sweet Spots

Actionable Advice:

- **RAM:** 8GB minimum, 16GB optimal for most users, 32GB+ only for specialized use
- **Storage:** 512GB optimal, 1TB+ for power users
- Prioritize SSD over HDD for performance-critical tasks

7.1.4 Build Quality vs. Specifications

Key Finding: Build quality tier (series) independently influences price beyond raw specifications. Premium series (ThinkPad, XPS, MacBook Pro) command prices 20–40% higher than consumer series with similar specs.

Actionable Advice:

- Budget-conscious buyers: Consumer series (IdeaPad, Inspiron) offer best value
- Durability seekers: Business series (ThinkPad, Latitude) justify premium for build quality
- Status/premium seekers: Premium series (XPS, MacBook Pro) command highest prices

7.1.5 Regional and Wilaya Insights

Key Finding: Market concentration is extremely high:

- **Alger** alone accounts for 70.8% of all listings (11,521 listings)
- Top 3 wilayas (Alger, Oran, Blida) account for 76.7% of all listings
- **North Coast** region: 88.2% of listings, median price 98,000 DZD (+3.2% vs overall)

- **Highlands:** 10% of listings, median price 75,000 DZD (−21.1% vs overall)
- **South:** 1.3% of listings, median price 66,000 DZD (−30.5% vs overall)

Implication: Buyers in less-represented wilayas may face fewer local options; sellers in major wilayas benefit from higher visibility. Regional price discounts in Highlands and South regions may reflect different market dynamics or product mix.

Actionable Advice:

- Buyers: Consider nationwide listings if local supply is thin; regional medians suggest potential savings outside North Coast.
- Sellers: In high-concentration wilayas (especially Alger), competitive pricing and feature highlighting matter most.

7.1.6 Brand and Gaming Series Premiums

Brand and segment positioning (gaming vs. consumer) drive price significantly beyond raw specifications.

Actionable Advice:

- Buyers: For similar specs, consumer series (IdeaPad, Inspiron) offer lower prices than gaming (Legion, ROG); Apple commands a +44% brand premium even for similar specifications.
- Sellers: Highlight gaming series or premium brand when listing; use segment-based comparisons for competitive positioning.

7.2 Insights for Sellers

7.2.1 Competitive Pricing Strategy

Key Finding: The best model (Gradient Boosting) achieves MAPE of 15.17% on the temporal test set. Sellers can use model predictions as pricing baselines, adjusting for:

- Condition (apply appropriate depreciation)
- Market timing (back-to-school premium)
- Unique features (highlight premium features)

Actionable Advice:

- Price within $\pm 10\%$ of model prediction for competitive positioning
- Highlight high-value features (CPU generation, RAM, SSD) in listings

7.2.2 Temporal Pricing Opportunities

Key Finding: Back-to-school period (August–October) shows increased listing volume and potential price premiums. Thus; sellers can time listings to maximize visibility and potentially command higher prices.

Actionable Advice:

- List during back-to-school period for maximum visibility
- Consider slight price premium (5–10%) during high-demand periods

8 Association Rules Exploration

We ran an exploratory association-rule analysis on the same preprocessed data (Apriori with lift-based ranking) to see whether co-occurrence patterns could reveal hidden structure beyond regression and clustering. Numeric features (price, RAM, storage, CPU/GPU tiers) were binned into binary items; we mined frequent itemsets with a minimum support threshold and derived rules ranked by **lift** (positive association only).

8.1 Ideal Scenario vs. Laptop Data Reality

Association rule mining is designed for *market-basket data*, where multiple items co-occur independently within the same transaction. In our dataset, each observation is a *single laptop listing*: CPU, GPU, RAM, storage, and price are jointly determined by product design and listing structure. Thus we are mining **product design / configuration co-occurrence**, not **customer purchasing behavior**. The following contrast justifies why association rules are a poor fit:

- **Ideal**: Transaction data (shopping baskets); binary presence/absence; independent item selection; large itemset variety; customer choice data.
- **Reality**: Single-item listings (one laptop per row); continuous specs discretized (information loss); specs highly correlated by design; limited configuration combinations; seller listing data (supply, not demand).

8.2 Consequences and Interpretation

As a result:

- **Redundant rules**: Many rules are overlapping subsets of the same configuration, producing a large number of near-duplicate rules.
- **Artificially high lift**: High lift arises from variables derived from the same listing rather than meaningful co-occurrence.
- **Loss of information**: Discretization of price and performance scores creates trivial antecedent–consequent relationships from the same underlying signals.

Rules discovered should be interpreted as “laptops with X tend to also have Y” (product design / listing structure), not “customers who want X also want Y.” Association rules are therefore ill-suited as a primary tool for this problem; predictive modeling and segmentation provide more appropriate and informative frameworks. The exploratory run still informs configuration co-occurrence and pricing-strategy context where useful.

9 Limitations and Future Work

9.0.1 Limitations

- **Data Quality**: High missingness in some features (GPU, screen resolution) limits their utilization.

- **Temporal Stability:** Model assumes stable relationships; market shifts may require periodic retraining.
- **High-Price Accuracy:** Limited training data for laptops > 600,000 DZD reduces accuracy at the high end.
- **Feature Interactions:** Some complex interactions may not be fully captured by current features.
- **External Factors:** Market dynamics (supply/demand, economic conditions) not explicitly modeled.

9.0.2 Future Work

1. Enhanced Feature Engineering:

- Incorporate screen resolution and panel type (IPS, OLED) when available
- Add battery life estimates based on CPU power class and capacity
- Include warranty information and seller reputation scores

2. Market Analysis:

- Analyze price trends over time to identify market cycles
- Study regional price variations (city-level analysis)
- Investigate brand-specific depreciation patterns